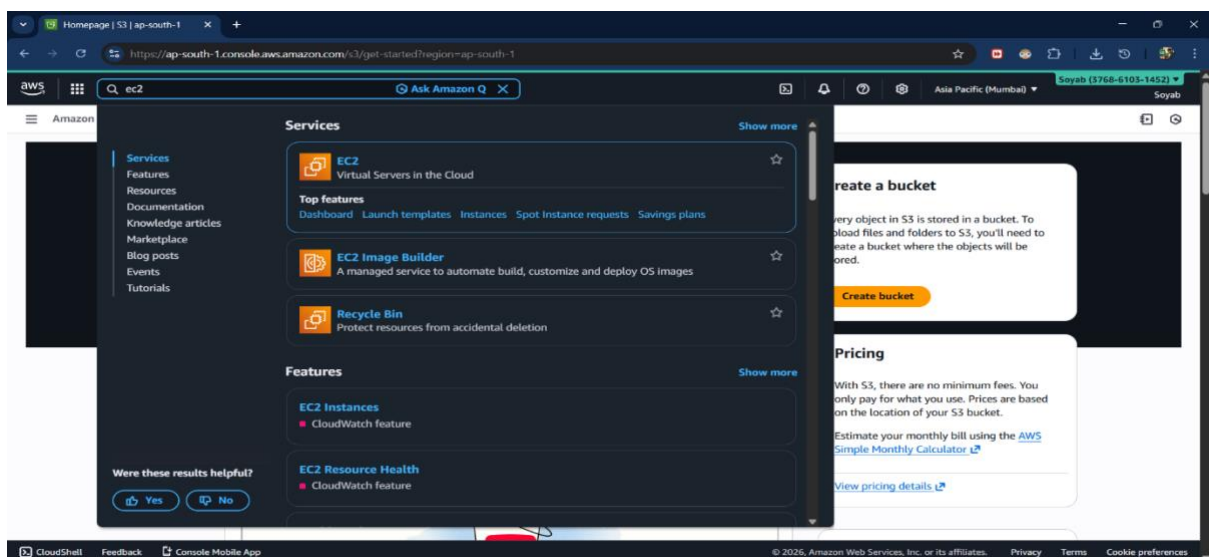


NAME:Md Soyab EXAM ROLL-11600223089 CLASS ROLL-CSE/23/112

Problem Statement: Hosting a Website on EC2.

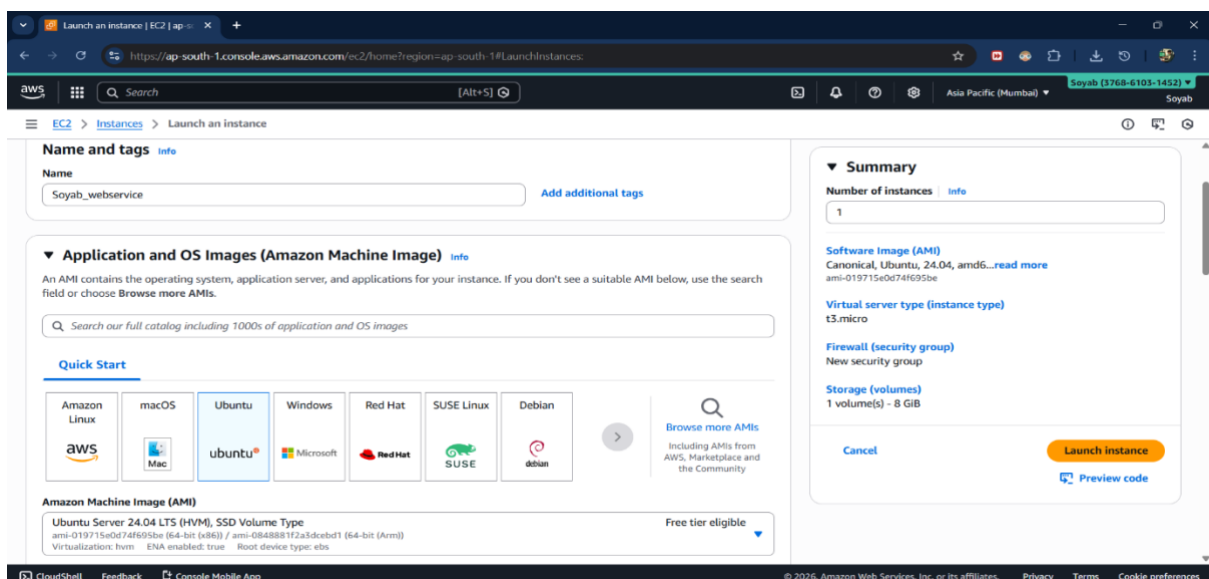
Step 1: Open EC2 Service

Log in to the AWS Management Console. Search for **EC2** in the top search bar and open the **Amazon EC2** service. This service allows you to create and manage virtual servers.



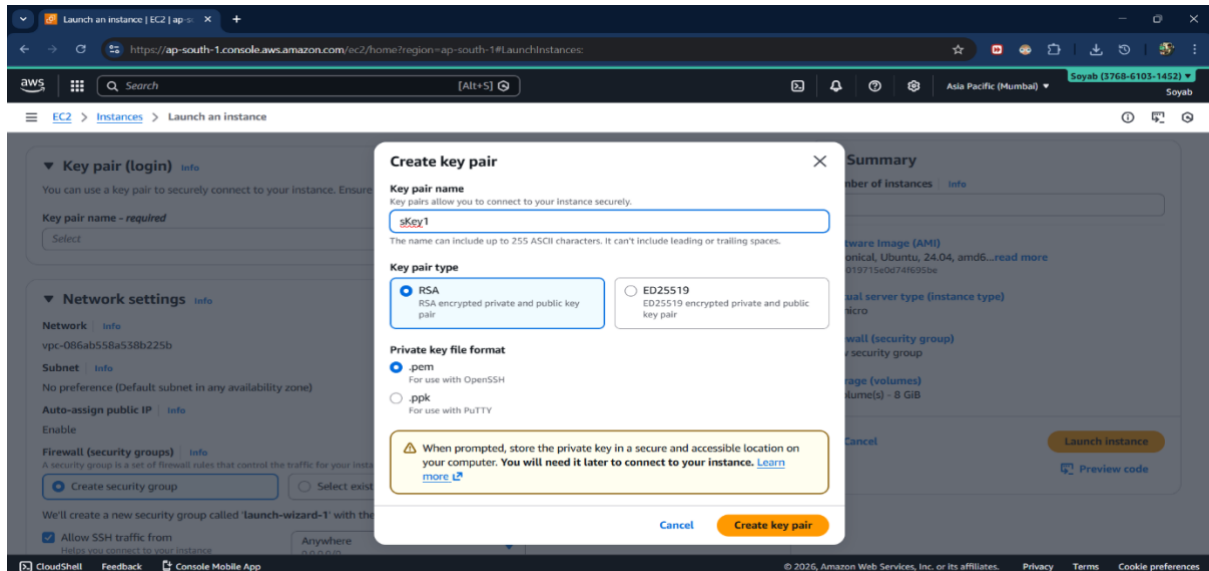
Step 2: Launch a New EC2 Instance

Click **Launch Instance**. Provide a name for the instance (Ec2-Instance). Choose an **Amazon Machine Image (AMI)** such as **Amazon Ubuntu**. Select an **Instance Type** (t3.micro, free-tier eligible).



Step 3: Create or Select a Key Pair

Under **Key pair**, select an existing key pair or create a new one. Download the key pair file (key2.pem) and keep it safe. This key is required to connect to the EC2 instance securely.



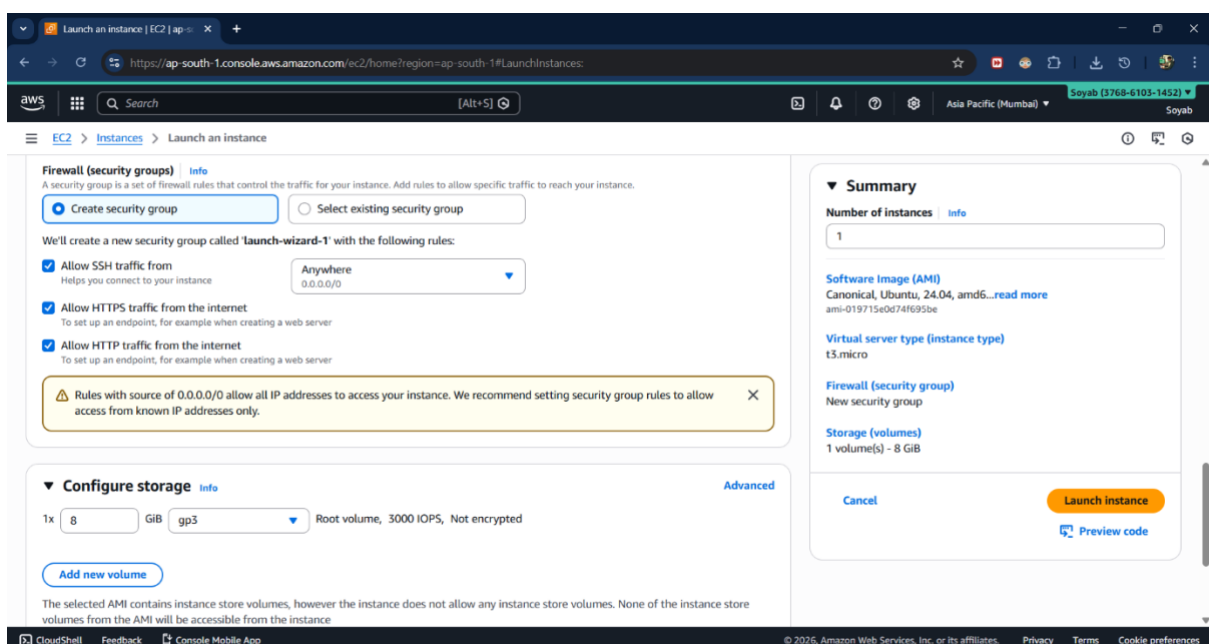
Step 4: Configure Security Group

Create a new security group and allow the following inbound rules:

- SSH (Port 22) – for remote login

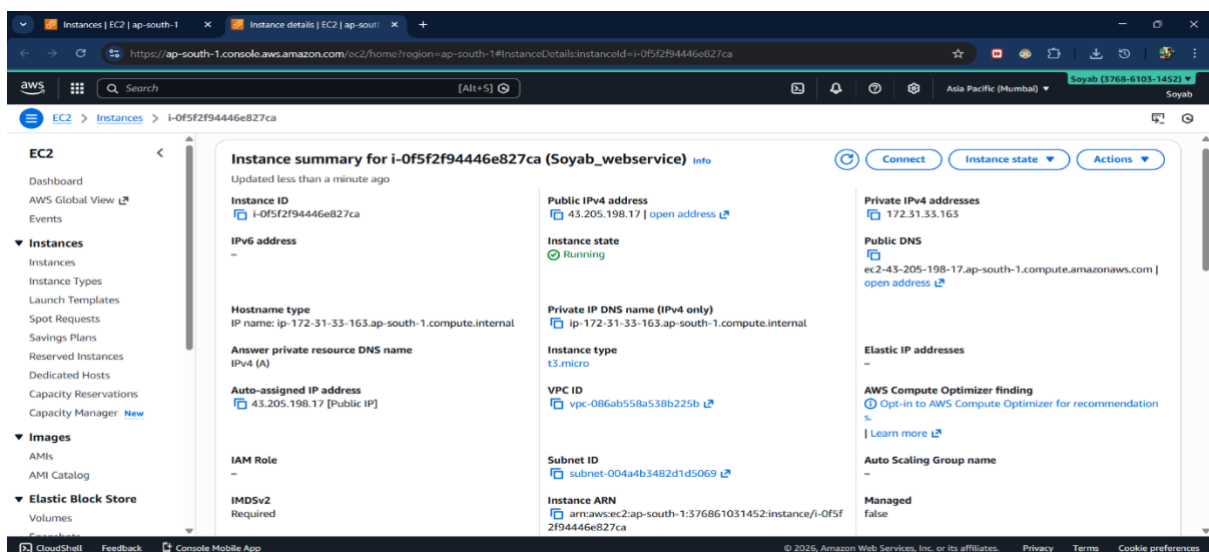
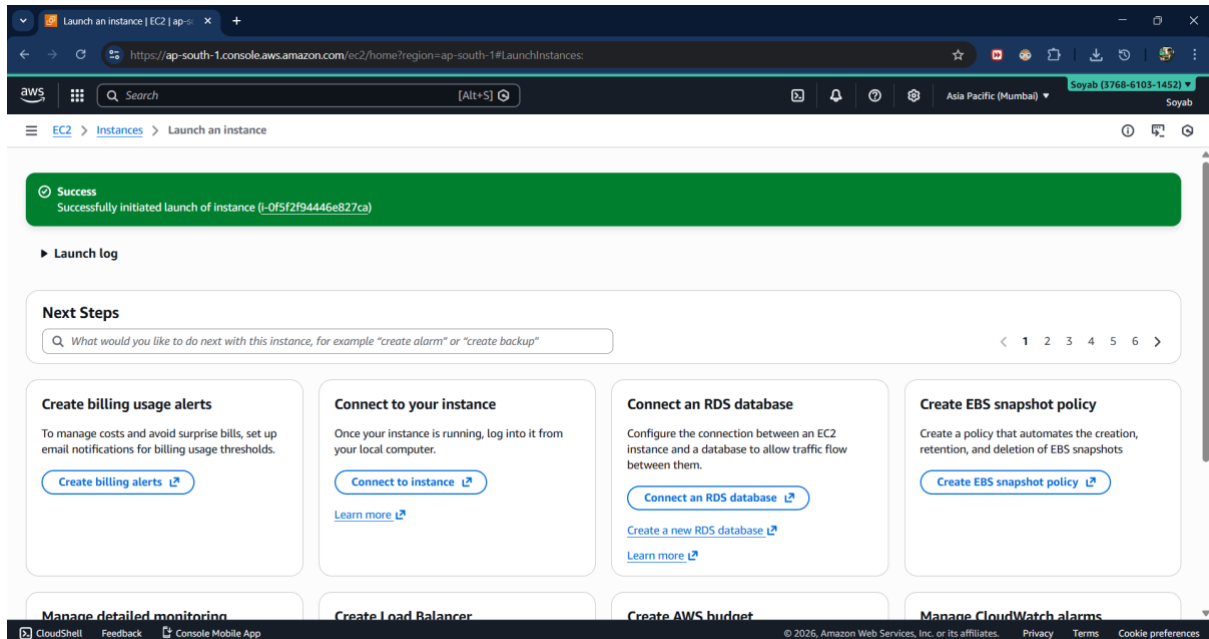
- HTTP (Port 80) – for website access

Source can be set to **Anywhere (0.0.0.0/0)**.



Step 5: Launch the Instance

Review all settings and click **Launch Instance**. Wait until the instance state changes to **Running**.

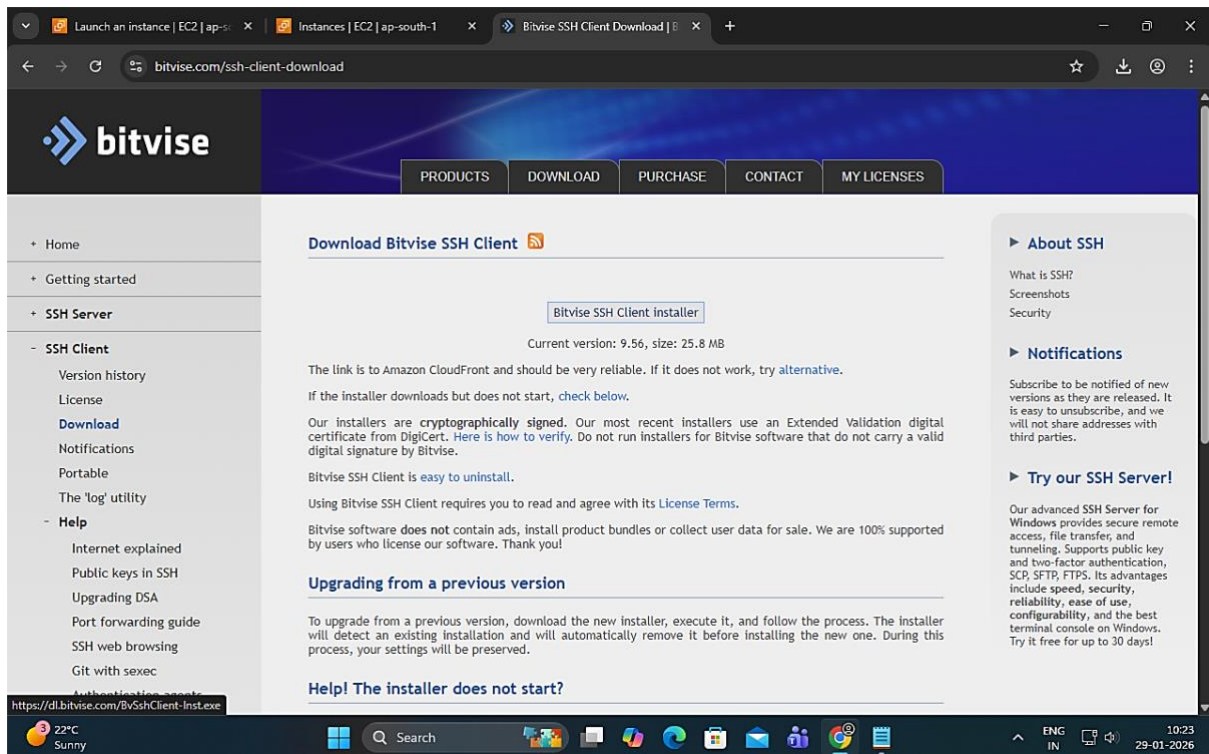


Step 6: Connect to the EC2 Instance Using Bitwise SSH Client (v9.56)

Bitwise SSH Client is a GUI-based tool used to securely connect to an EC2 instance using SSH.

6.1 Download and Install Bitwise

Open a browser and go to the official **Bitwise** website. Download **Bitwise SSH Client (v9.56)** and install it on your system.



Step 6.2: Configure Bitvise and Load Key Pair

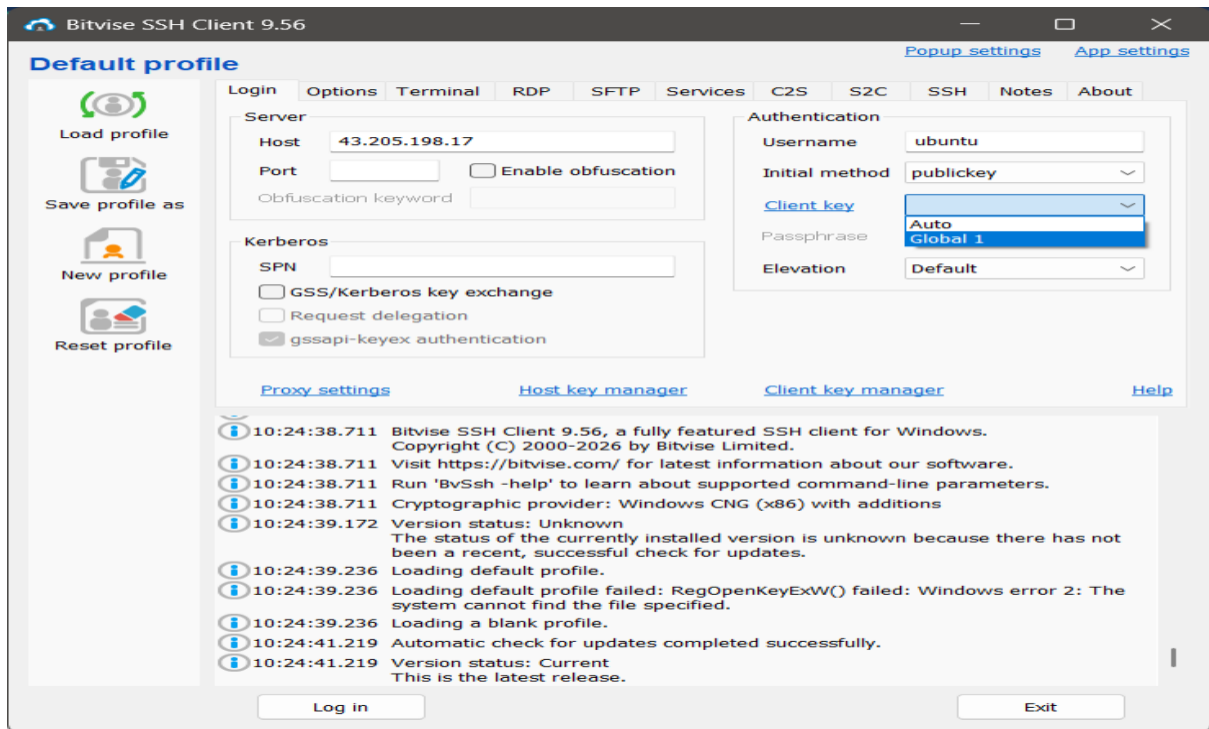
Open Bitvise SSH Client and in the Login tab enter:

.Host: Public IPv4 address (e.g., 13.233.200.152)

.Port: 22

.Username: ubuntu

Under Authentication, select Publickey, click Client key manager, import the .pem key file, click OK, and ensure the key is selected.



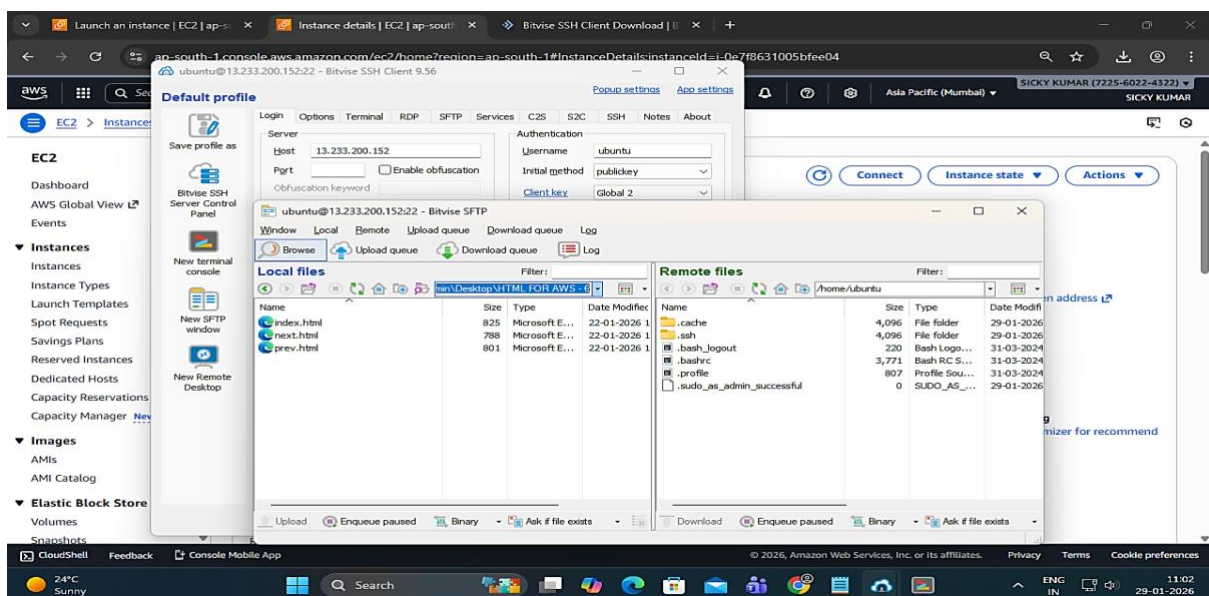
6.3 Connect to EC2

Click **Login**.

Accept the host key when prompted.

The terminal window opens.

EC2 instance is successfully connected using Bitvise SSH Client.



Step 7: Install and Configure Nginx Web Server

After connecting to the EC2 instance, install and start **Nginx**, which is a lightweight and high-performance web server.

1) Update the system packages:

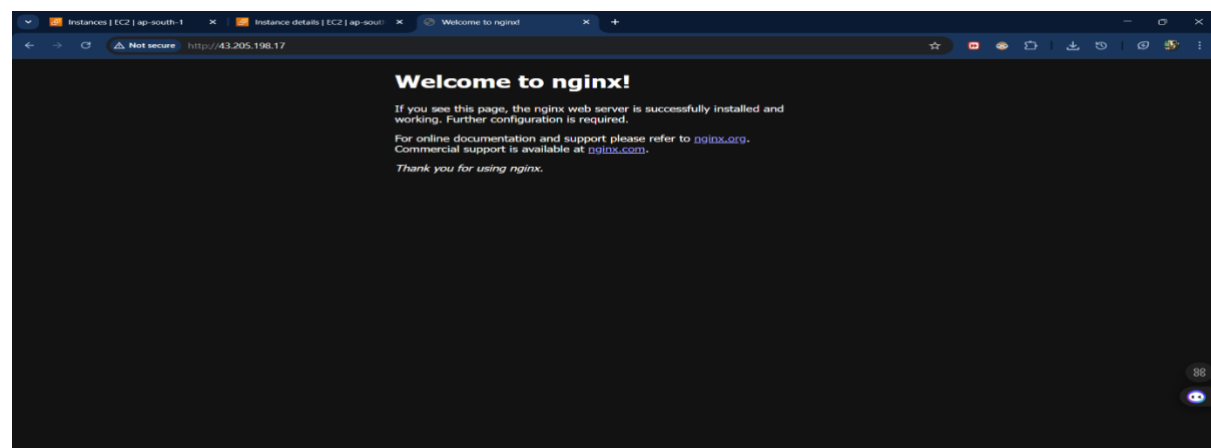
```
sudo apt upgrade -y
```

2) Install Nginx:

```
sudo apt install nginx -y
```

```
ubuntu@43.205.198.17:22 - Bitvise xterm - ubuntu@ip-172-31-33-163: ~
Get:33 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [29.5
kB]
Get:34 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [17.9
kB]
Get:35 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10
.5 kB]
Get:36 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata
[1444 B]
Get:37 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [
216 B]
Get:38 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 c-n-f Metada
ta [116 B]
Get:39 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [
212 B]
Get:40 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 c-n-f Metada
ta [116 B]
Get:41 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1410 kB]
Get:42 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [229 kB]
Get:43 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:44 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [9820 B]
Get:45 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [925 kB]
Get:46 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [209 kB]
Get:47 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:48 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [19.7 kB]
Get:49 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2369 kB]
Get:50 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [543 kB]
Get:51 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:52 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [536 B]
Get:53 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [28.8 kB]
Get:54 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [6492 B]
Get:55 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:56 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Fetched 39.5 MB in 6s (6229 kB/s)
Reading package lists... Done
ubuntu@ip-172-31-33-163:~$
```

3) Verify Nginx is running by opening the EC2 Public IPv4 address in a browser.



Step 8: Give Directory Permission and Upload Website Files Using Bitvise.

After connecting to the EC2 instance using Bitvise SSH Client:

1.Open the Terminal in Bitvise and give permission to the web directory:

```
cd /var/www
```

```
sudo chmod 777 html
```

2.Open the SFTP window in Bitvise.

3.On the Remote side, go to:

```
/var/www/html
```

4.From the Local side, select three HTML files and drag and drop them into the remote directory.

5.Refresh the browser using the EC2 Public IPv4 address to view the uploaded pages.

6.Permissions are set first then website files are uploaded successfully .

